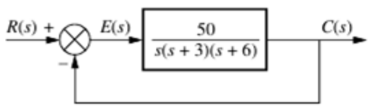
University of Cape Town

Department of Electrical Engineering

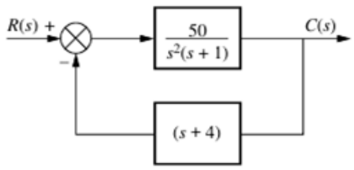
EEE3094S - Control Systems Engineering

Problem Set 5.2: Closed Loop Stability Analysis

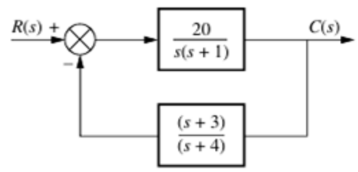
Question 1



System 1

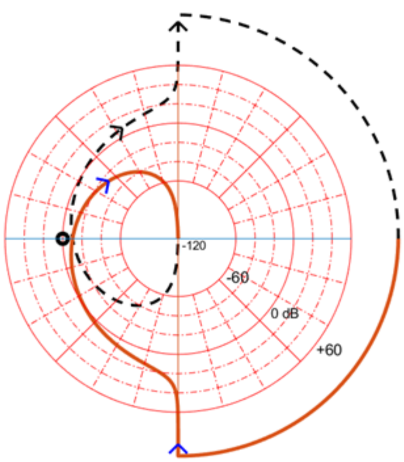


System 2

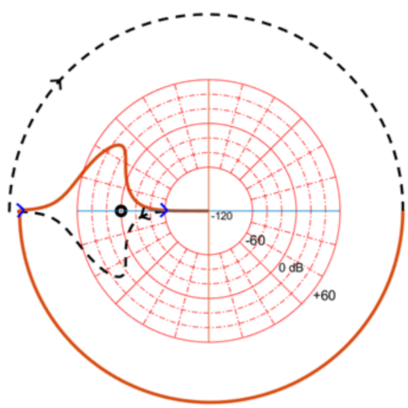


System 3

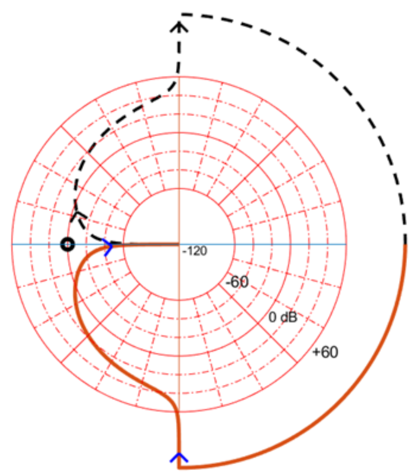
The systems above have the following Nyquist plots: (Black circle = -1)



(a) System 1



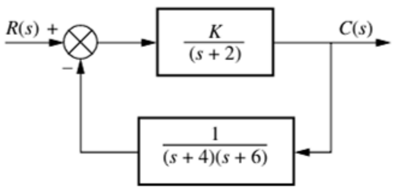
(b) System 2



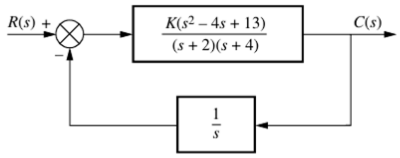
(c) System 3

Using the Nyquist stability criterion, determine whether each system is stable.

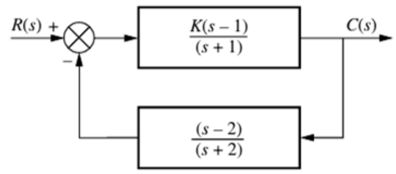
Question 2



System 1

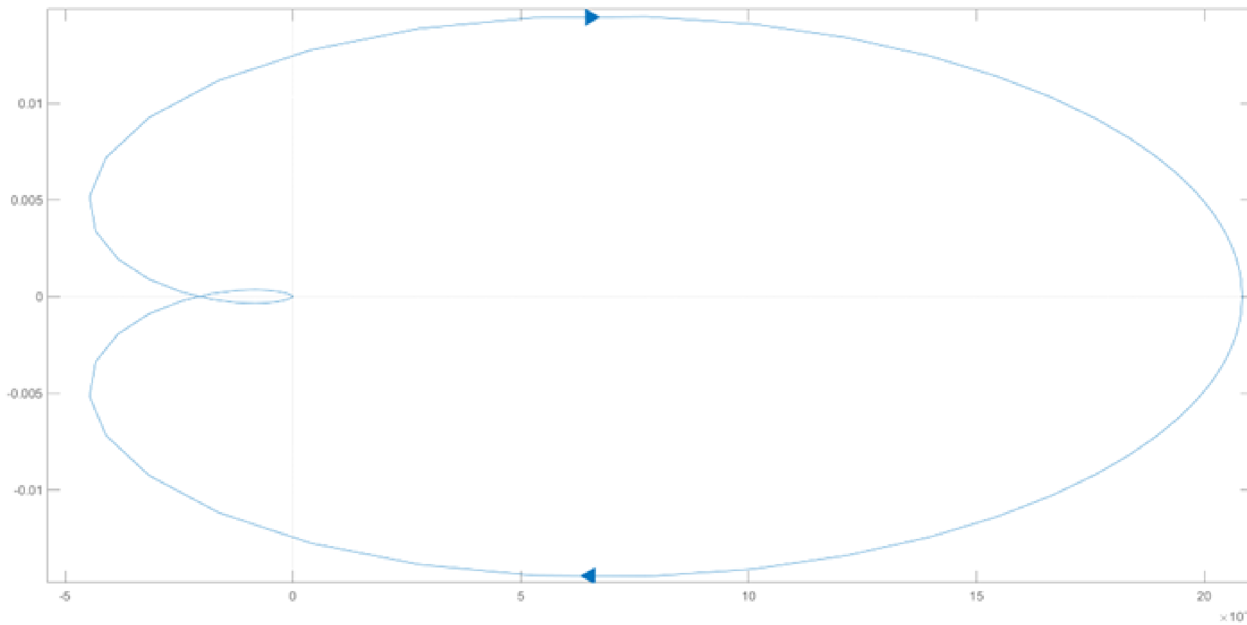


System 2

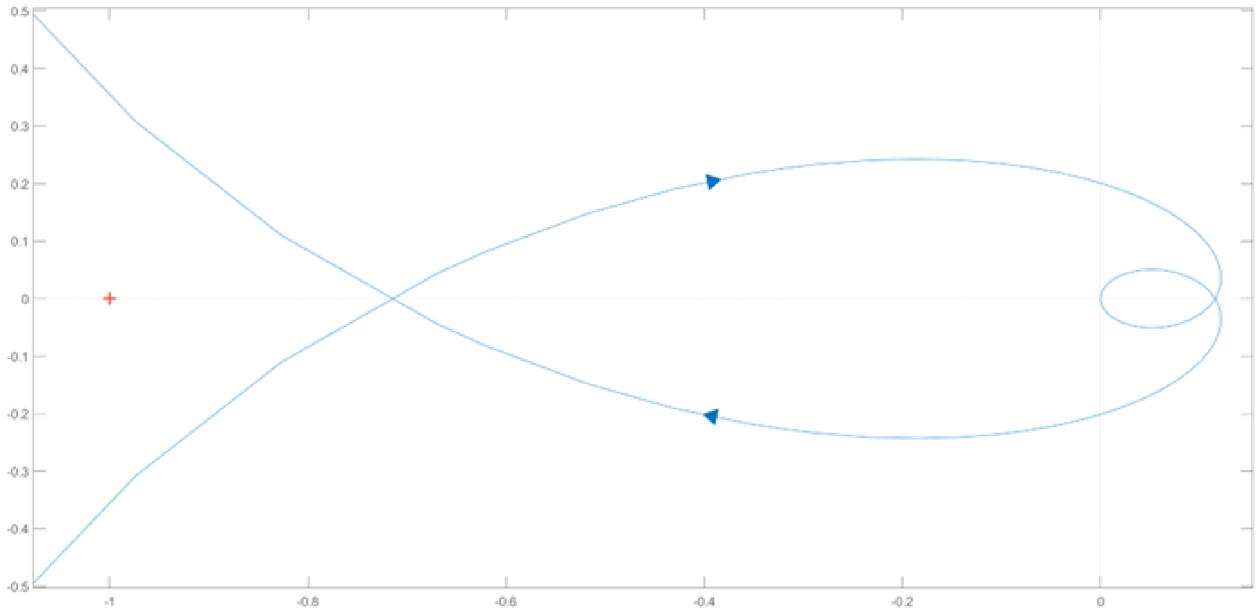


System 3

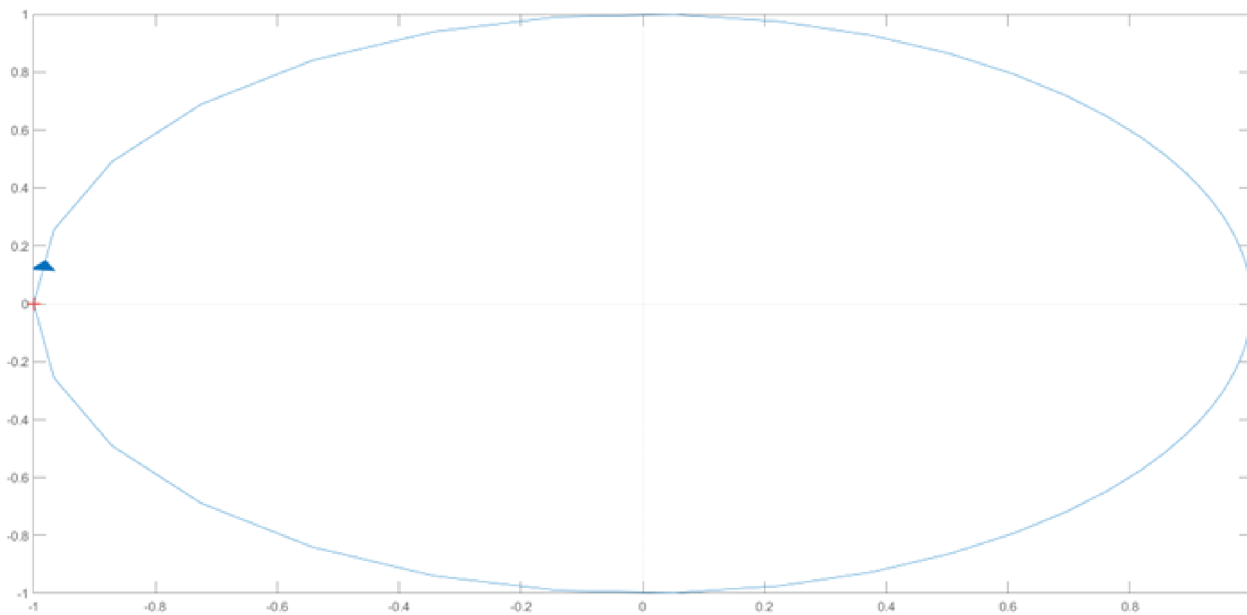
The systems above have the following Nyquist plots for unity gain ($K = 1$):



System 1



System 2

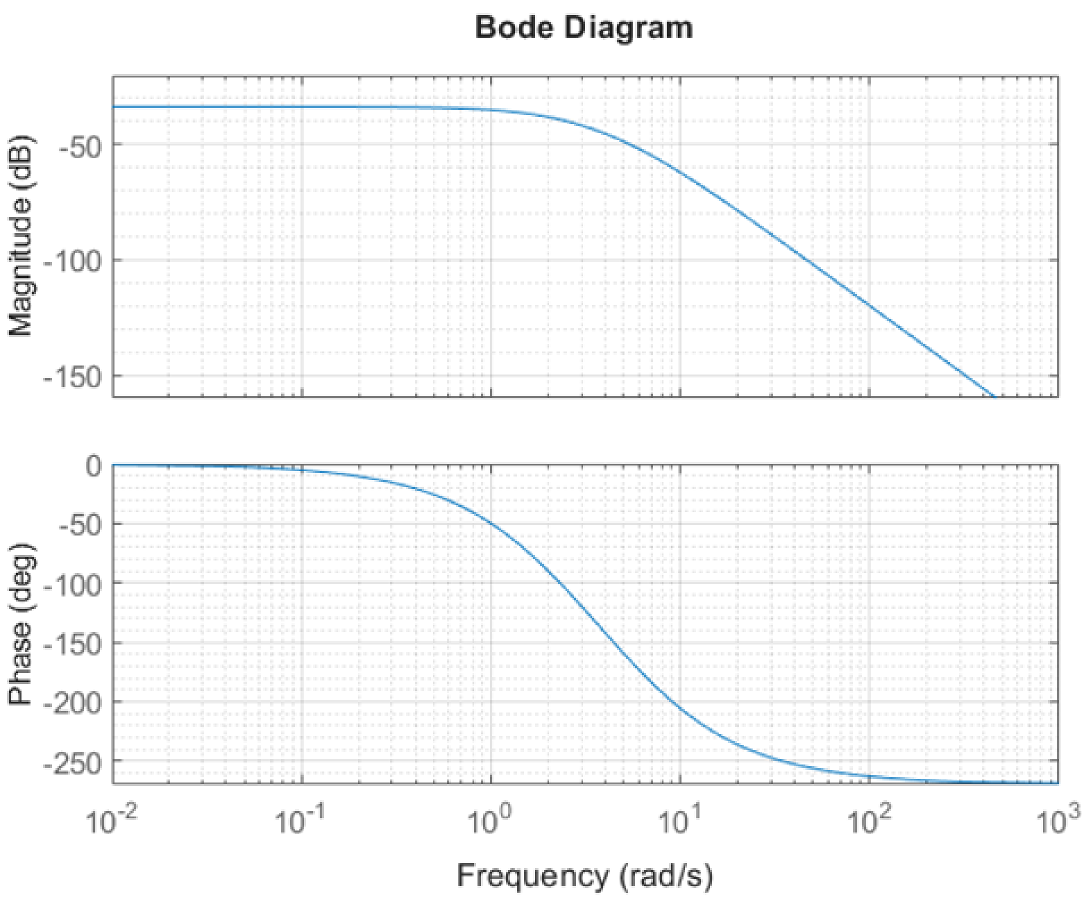


System 3

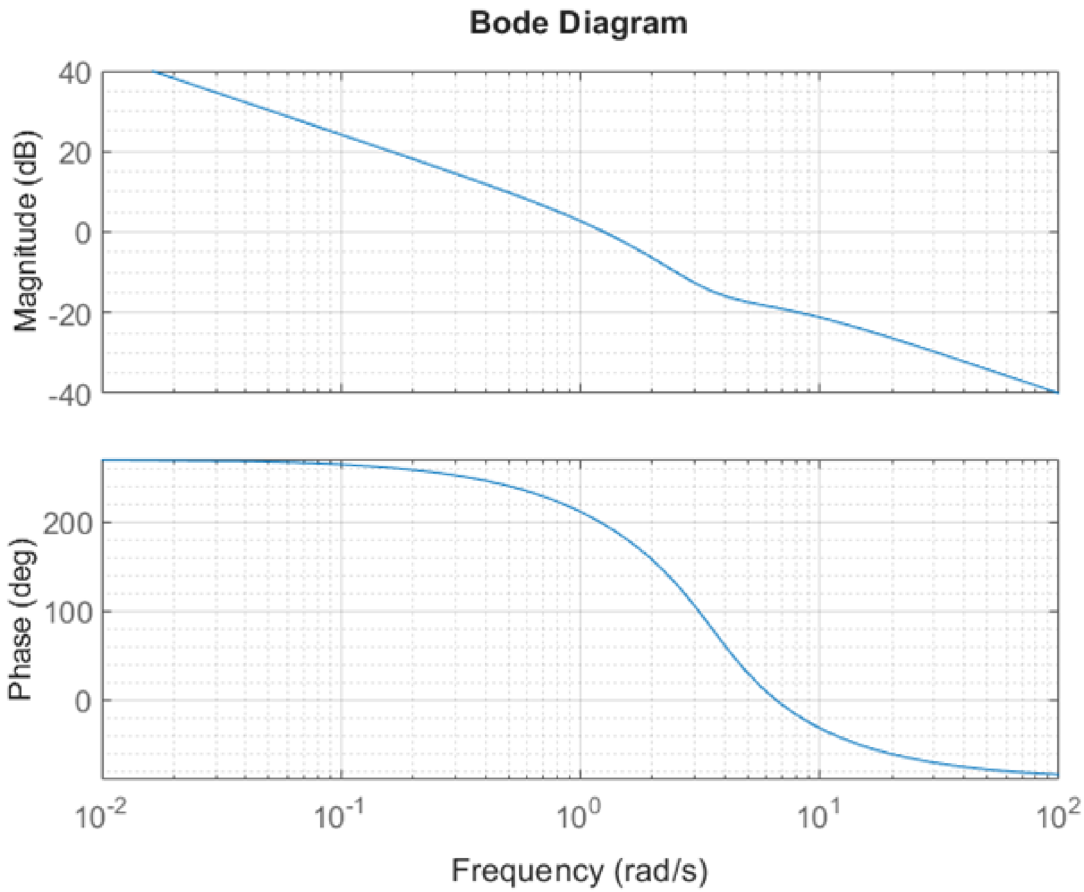
Using the Nyquist stability criterion, find the range of K values for which each system is stable.

Question 3

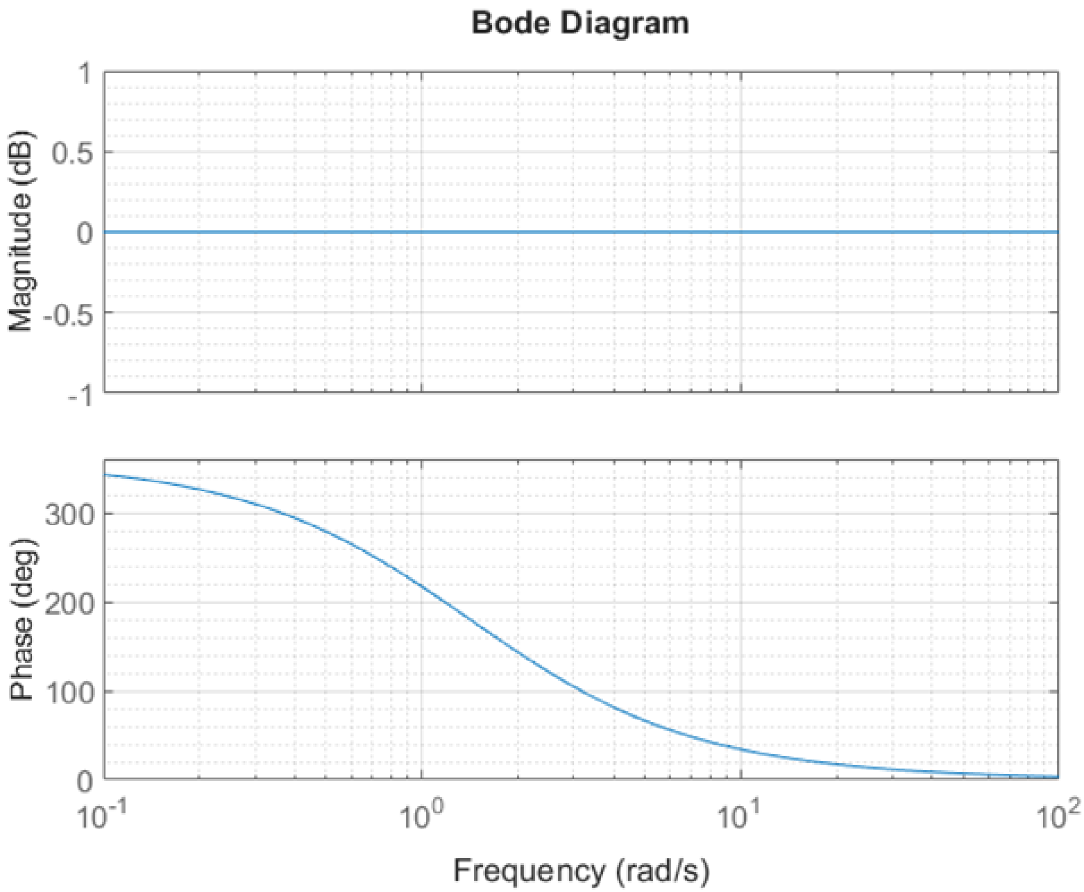
The systems in Question 2 have the following open-loop frequency responses for unity gain:



System 1


Bode Diagram
40
20
0
-20
-40
Magnitude (dB)
200
100
0
Phase (deg)
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Frequency (rad/s)

System 2



System 3

For each system in Question 2, find the gain margin and phase margin if the value of K is:

i) $K = 1000$

ii) $K = 100$

iii) $K = 0.1$

Questions adapted from Problems 9–11 in Nise, *Control Systems Engineering*, Chapter 10.